

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I Pugazharasan K (192511341) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Pugazharasan K

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Cloud Computing and Big Data Analytics

- 1.) Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Definition: cloud computing is the delivery of computing services such as servers, storage, databases, networking, software and analytics over the Internet on a pay-as-you-use basis.

Characteristics:

- 1.) On-Demand self-service: Users can access computing resources whenever needed without human intervention.
- 2.) Broad network access: Services are available over the Internet and can be accessed from computers, laptops.
- 3.) Resource pooling: Resources are shared among multiple users using virtualization, improving efficiency.
- 4.) Rapid elasticity: Resources can be increased or decreased quickly based on demand.

- 2.) Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

- (i) Hardware evolution: Computers evolved from mainframes to powerful servers with increased processing power and storage.
- (ii) Virtualization: One physical server could run multiple virtual machines, improving resource utilization.
- (iii) Internet evolution: High-speed Internet enabled remote access to applications and data.
- (iv) Web-based software: Applications become browser-based (SaaS) allowing users to access software online.

(v) Major cloud platforms: Technologies like virtualization, automation and distributed data centers led to cloud platforms such as AWS, Microsoft Azure, and Google Cloud.

3) Explain the role of server virtualization in cloud computing. Diff. virtualization as an enabling technology and cloud computing as a service model.

Role of server virtualization:-

- Creates multiple virtual machines on a single physical server.
- Improves hardware utilization.
- Reduces infrastructure costs.
- Makes resource allocation flexible and efficient.

Virtualization	cloud computing
(i) Technology that creates virtual machines.	(i) Service model that delivers IT resources over the Internet.
(ii) Focuses on efficient hardware use.	(ii) Focuses on providing services (IaaS, PaaS, SaaS).
(iii) Runs on a physical server.	(iii) Uses virtualization to provide cloud services.

4) Discuss the role of data centers in delivering cloud services.

Data centers are facilities that contain servers, storage devices, networking equipment, power systems, and cooling systems. They are the backbone of cloud computing.

Roles:-

- Store and process user data.
- Host cloud applications and services.
- Provide high availability and reliability.
- Ensure security through firewalls and backups.

- Support disaster recovery and business continuity.
- Enable scalable cloud services.

5.) Differentiate IaaS, PaaS, SaaS and XaaS with one suitable real-world example for each model.

Service Model	Description	Example
IaaS (Infrastructure as a service)	Provides virtual servers, storage and networking.	Amazon EC2 (AWS).
PaaS (Platform as a service)	Provides a platform to develop and deploy applications.	Google app engine.
SaaS (Software as a service)	Delivers software applications over the Internet.	Google docs.
XaaS (Anything as a service)	Offers any IT service through the cloud.	Microsoft 365.